PART A- FOUNDATIONS OF MACHINE LEARNING

A1

- Supervised Learning is a ML approach where a model learns from data that contains known target labels. The model learns the relationship between input features and the target so it can make predictions on new data. While Unsupervised Learning uses data without target labels and attempts to discover hidden patterns in the data.
- For the Diabetes dataset Supervised Learning is used for regression because the target represents the progression after one year, which is a numerical value. For the Ionosphere dataset Supervised Learning is used for classification because the target identifies radar returns as either good or bad. For the Wholesale Customer dataset, unsupervised learning is used for clustering because the features are used to group customers according to similarities without using labels.

A2

- The **bias-variance trade-off** describes the balance between two sources of prediction error. **Bias** is the error caused by a model being too simple and making strong assumptions about the data. **Variance** is the error caused by a model being too sensitive to the training data. A good model should have an appropriate balance between the two.
- For the Diabetes regression dataset, I expect the **Decision Tree Regressor** to have **lower bias but higher variance** than Linear Regression. A decision tree can learn complex, non-linear relationships, allowing it to fit the training data more closely. However, this flexibility can make it sensitive to changes in the training data. Linear Regression is simpler and therefore generally has higher bias but lower variance.

A3

- **Overfitting** occurs when a model learns the training data too closely, including noise, and therefore performs poorly on unseen data. **Underfitting** occurs when a model is too simple to capture the important patterns in the data, resulting in poor performance on both training and test data.
- Overfitting can be detected by **comparing training and test performance** and by using **5-fold cross-validation** to determine whether performance remains consistent on different subsets of the data.
- Overfitting can be reduced by **limiting the complexity of the Decision Tree**, for example by setting its maximum depth, and by using **cross-validation** to select suitable model settings

A4

- Accuracy alone may be misleading for the **Ionosphere** dataset because it only measures the overall proportion of correctly classified observations. If one class is more common than the other, a model could achieve high accuracy while performing poorly on the less common class.

B1- SIMPLE LINEAR REGRESSION

B1 (a) Calculate the slope.

x	y	x^2	xy
1	10	1	10
2	9	4	18
3	14	9	42
4	11	16	44
5	16	25	80
$\Sigma = 15$	$\Sigma = 60$	$\Sigma x^2 = 55$	$\Sigma xy = 194$

$$\text{Slope } (m) = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

$$m = \frac{5(194) - (15)(60)}{5(55) - (15)^2}$$

$$m = \frac{970 - 900}{275 - 225} = \frac{70}{50}$$

$$m = 1.400$$

$$\text{Intercept } c = \frac{\Sigma y - m\Sigma x}{n}$$

$$c = \frac{60 - 1.4(15)}{5} = \frac{39}{5}$$

$$c = 7.800$$

Equation $\Rightarrow \hat{y} = 1.400x + 7.800$

b)

Predicted formula $\Rightarrow \hat{y} = 1.4x + 7.8$

Residual $\Rightarrow e = y - \hat{y}$

x	Actual y	Predicted $\hat{y}$	Residual $y - \hat{y}$
1	10	9.200	0.800
2	9	10.600	-1.600
3	14	12.000	2.000
4	11	13.400	-2.400
5	16	14.800	1.200

$x = 1$

$y = 1.4(1) + 7.8 = 9.2$

$e = 10 - 9.2 = 0.8$

Predictions $\Rightarrow$ 9.200, 10.600, 12.000, 13.400, 14.800

Residuals $\Rightarrow$ 0.800, -1.600, 2.000, -2.400, 1.200

c) Mean Squared Error (MSE)

$$MSE = \frac{\sum (y - \hat{y})^2}{n}$$

Residual	Squard R.
0.8	0.64
-1.6	2.56
2.0	4.00
-2.4	5.76
1.2	1.44
	$\sum e^2 = 14.40$

$$MSE = \frac{14.40}{5}$$

$$= 2.880$$

d) Coefficient of Determination R^2

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Mean of $y \Rightarrow \bar{y} = \frac{60}{5} = 12$

$SS_{tot} \Rightarrow \sum (y - \bar{y})^2$

$= (10-12)^2 + (9-12)^2 + (14-12)^2 + (11-12)^2 + (16-12)^2$

$= 34$

$$R^2 = 1 - \frac{14.4}{34}$$

$$R^2 = 1 - 0.423529$$

$$R^2 = 0.576$$

B2. K - Nearest Neighbour. (KNN)

Q = (3, 2)

$T1 \Rightarrow d(Q, T1) = \sqrt{(3-3)^2 + (2-7)^2}$

$= 5$

$T2 \Rightarrow d(Q, T2) = \sqrt{(3-8)^2 + (2-1)^2}$

$= 5.10$

$T3 \Rightarrow d(Q, T3) = \sqrt{(3-9)^2 + (2-1)^2}$

$= 6.08$

$T4 \Rightarrow d(Q, T4) = \sqrt{(3-9)^2 + (2-8)^2}$

$= 7.21$

$T5 \Rightarrow d(Q, T5) = \sqrt{(3-9)^2 + (2-8)^2}$

$= 7.81$

$T6 \Rightarrow d(Q, T6) = \sqrt{(3-1)^2 + (2-1)^2}$

$= 2.24$

Point	Class	Distance from Q
T6	B	2.24
T1	A	5.00
T2	B	5.10
T3	A	6.08
T4	A	7.21
T5	B	7.81

b) Classification using K = 3

T6 → B Votes ⇒ Class A = 1

T1 → A Class B = 2

T2 → B

Q → B

c) Classification using K = 1

Nearest point = T6	⇒ Both predictions are class B
Class = B	⇒ For this particular point changing k from 1 to 3 does not change the prediction
Q → B	
K = 3 → B	
K = 1 → B	

B3. One Iteration of K-Means Clustering

K = 2 $C_1 = (1, 10)$ $C_2 = (8, 8)$

$d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2$

P1 = (5, 1) to $C_1 = (1, 10) \Rightarrow (5-1)^2 + (1-10)^2$

$= 97$

To $C_2 = (8, 2) \Rightarrow (5-8)^2 + (1-8)^2$

$= 58$

P1 → C_2

P2 = (1, 9) To $C_1 \Rightarrow (1-1)^2 + (9-10)^2 = 1$

To $C_2 \Rightarrow (1-8)^2 + (9-8) = 50$

P2 → C_1

P3 = (3, 2) To $C_1 \Rightarrow (3-1)^2 + (2-10)^2 = 8$

To $C_2 \Rightarrow (3-8)^2 + (2-8)^2 = 25$

P3 → C_1

P4 = (8, 8) To $C_1 \Rightarrow (8-1)^2 + (8-10)^2 = 53$

To $C_2 \Rightarrow (8-8)^2 + (8-8)^2 = 0$

P4 → C_2

P5 = (9, 3) To $C_1 \Rightarrow (9-1)^2 + (3-10)^2 = 113$

To $C_2 \Rightarrow (9-8)^2 + (3-8)^2 = 26$

P5 → C_2

P6 = (1, 10) To $C_1 \Rightarrow (1-1)^2 + (10-10)^2 = 0$

To $C_2 \Rightarrow (1-8)^2 + (10-8)^2 = 53$

P6 → C_1

Cluster 1 ⇒ $C_1 = \{P2, P3, P6\}$

Cluster 2 ⇒ $C_2 = \{P1, P4, P5\}$

Part D – Reflection

This assignment has improved my understanding of how different machine learning techniques are applied to real datasets. I learned the difference between supervised and unsupervised learning and how regression, classification, and clustering solve different types of problems. Through the practical work, I gained experience in exploratory data analysis, preprocessing, model training, evaluation, cross-validation, and visualization. I also learned that selecting a machine learning model should be based on its performance rather than simply choosing a more complex algorithm.

One challenge I encountered was obtaining poor performance from the Decision Tree Regressor on the Diabetes dataset. The model produced a negative R^2 score on the test set and also performed poorly during cross-validation. I addressed this by comparing its MAE, RMSE, and R^2 values with those of Linear Regression and using 5-fold cross-validation to evaluate generalization. Linear Regression performed better, so it was selected as the better model for this dataset. This showed me the importance of testing models carefully rather than assuming that a more complex model will always perform better.

An important consideration in machine learning is the potential for biased or unfair decisions. If a dataset contains biased or unrepresentative information, a model may produce unreliable results for certain groups. Therefore, models should be evaluated using appropriate metrics, and their results should not be used without considering their possible impact on people.

The assignment strengthened my practical machine learning skills and showed me the importance of reproducible analysis, clear visualizations and appropriate evaluation metrics.

b) Compute the new centroids.

New C_1

$P_2 = (1,9)$, $P_3 = (3,8)$ $P_5 = (1,10)$

Mean of $x_1 = \frac{1+3+1}{3}$

$= 1.667$

Mean of $x_2 = \frac{9+8+10}{3}$

$= 9$

$C_1' = (1.667, 9.000)$

New C_2

$P_1 = (5,1)$ $P_4 = (8,2)$ $P_5 = (9,3)$

Mean of $x_1 = \frac{5+8+9}{3} = 7.333$

Mean of $x_2 = \frac{1+2+3}{3} = 4$

$C_2' = (7.333, 4.000)$

c) Within-Cluster Sum of Squares (Inertia)

Cluster 1

$C_1' = (1.667, 9)$

$P_2 = (1,9) \Rightarrow (1-1.667)^2 + (9-9)^2$

$= 0.444$

$P_5 = (1,10) \Rightarrow (1-1.667)^2 + (10-9)^2$

$= 0.444 + 1$

$= 1.444$

$P_3 = (3,8) \Rightarrow (3-1.667)^2 + (8-9)^2$

$= 1.778 + 1$

$= 2.778$

$W_1 = 0.444 + 2.778 + 1.444$

$= 4.666$

Cluster 2

$C_2' = (7.333, 4.000)$

$P_1 = (5,1) \Rightarrow (5-7.333)^2 + (1-4)^2$

$= 5.444 + 9$

$= 14.444$

$P_4 = (8,8) \Rightarrow (8-7.333)^2 + (8-4)^2$

$= 0.444 + 16$

$= 16.444$

$P_5 = (9,3) \Rightarrow (9-7.333)^2 + (3-4)^2$

$= 2.778 + 1$

$= 3.778$

$W_2 = 14.444 + 16.444 + 3.778$

$= 34.666$

Total Inertia $= 4.666 + 34.666$

$= 39.332$